library load

```
In [86]: import pandas as pd
         import seaborn as sns
         import matplotlib.pyplot as plt
         import numpy as np
```

Datasets load

```
In [87]:
         df = pd.read_csv('HDHI Admission data.csv')
```

```
In [88]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15757 entries, 0 to 15756
Data columns (total 56 columns):
 #   Column                            Non-Null Count  Dtype 
---  ------                            --------------  ----- 
 0   SNO                               15757 non-null  int64 
 1   MRD No.                           15757 non-null  object
 2   D.O.A                             15757 non-null  object
 3   D.O.D                             15757 non-null  object
 4   AGE                               15757 non-null  int64 
 5   GENDER                            15757 non-null  object
 6   RURAL                             15757 non-null  object
 7   TYPE OF ADMISSION-EMERGENCY/OPD   15757 non-null  object
 8   month year                        15757 non-null  object
 9   DURATION OF STAY                  15757 non-null  int64 
 10  duration of intensive unit stay   15757 non-null  int64 
 11  OUTCOME                           15757 non-null  object
 12  SMOKING                           15757 non-null  int64 
 13  ALCOHOL                           15757 non-null  int64 
 14  DM                                15757 non-null  int64 
 15  HTN                               15757 non-null  int64 
 16  CAD                               15757 non-null  int64 
 17  PRIOR CMP                         15757 non-null  int64 
 18  CKD                               15757 non-null  int64 
 19  HB                                15505 non-null  object
 20  TLC                               15471 non-null  object
 21  PLATELETS                         15472 non-null  object
 22  GLUCOSE                           14894 non-null  object
 23  UREA                              15516 non-null  object
 24  CREATININE                        15510 non-null  object
 25  BNP                               7316 non-null   object
 26  RAISED CARDIAC ENZYMES            15757 non-null  int64 
 27  EF                                14252 non-null  object
 28  SEVERE ANAEMIA                    15757 non-null  int64 
 29  ANAEMIA                           15757 non-null  int64 
 30  STABLE ANGINA                     15757 non-null  int64 
 31  ACS                               15757 non-null  int64 
 32  STEMI                             15757 non-null  int64 
 33  ATYPICAL CHEST PAIN               15757 non-null  int64 
 34  HEART FAILURE                     15757 non-null  int64 
 35  HFREF                             15757 non-null  int64 
 36  HFNEF                             15757 non-null  int64 
 37  VALVULAR                          15757 non-null  int64 
 38  CHB                               15757 non-null  int64 
 39  SSS                               15757 non-null  int64 
 40  AKI                               15757 non-null  int64 
 41  CVA INFRACT                       15757 non-null  int64 
 42  CVA BLEED                         15757 non-null  int64 
 43  AF                                15757 non-null  int64 
 44  VT                                15757 non-null  int64 
 45  PSVT                              15757 non-null  int64 
 46  CONGENITAL                        15757 non-null  int64 
 47  UTI                               15757 non-null  int64 
 48  NEURO CARDIOGENIC SYNCOPE         15757 non-null  int64 
 49  ORTHOSTATIC                       15757 non-null  int64 
 50  INFECTIVE ENDOCARDITIS            15757 non-null  int64 
 51  DVT                               15757 non-null  int64 
 52  CARDIOGENIC SHOCK                 15757 non-null  int64 
 53  SHOCK                             15757 non-null  int64 
 54  PULMONARY EMBOLISM                15757 non-null  int64 
 55  CHEST INFECTION                   15757 non-null  object
dtypes: int64(39), object(17)
memory usage: 6.7+ MB
```

```
In [89]: df.duplicated()
```

```
Out[89]: 0        False
         1        False
         2        False
         3        False
         4        False
                  ...  
         15752    False
         15753    False
         15754    False
         15755    False
         15756    False
         Length: 15757, dtype: bool
```

Clean Column Names

```
In [90]:
         df.columns = df.columns.str.strip()
         df.head(10)
```

Out[90]:

	SNO	MRD No.	D.O.A	D.O.D	AGE	GENDER	RURAL	TYPE OF ADMISSION-EMERGENCY/OPD	month year	DURATION OF STAY	...	CONGENITAL	UTI	NEURO CARDIOGENIC SYNCOPE	ORTHOSTATIC	INFECTIVE ENDOCARDITIS	DVT	CARDIOGENIC SHOCK	SHOCK	PULMONARY EMBOLISM	CHEST INFECTION
0	1	234735	4/1/2017	4/3/2017	81	M	R	E	Apr-17	3	...	0	0	0	0	0	0	0	0	0	0
1	2	234696	4/1/2017	4/5/2017	65	M	R	E	Apr-17	5	...	0	0	0	0	0	0	0	0	0	0
2	3	234882	4/1/2017	4/3/2017	53	M	U	E	Apr-17	3	...	0	0	0	0	0	0	0	0	0	0
3	4	234635	4/1/2017	4/8/2017	67	F	U	E	Apr-17	8	...	0	0	0	0	0	0	0	0	0	0
4	5	234486	4/1/2017	4/23/2017	60	F	U	E	Apr-17	23	...	0	0	0	0	0	0	0	0	0	0
5	6	234675	4/1/2017	4/10/2017	44	M	U	E	Apr-17	10	...	0	0	0	0	0	0	0	0	0	0
6	7	234563	4/1/2017	4/6/2017	56	F	U	E	Apr-17	6	...	0	0	0	0	0	0	0	0	0	0
7	8	208455	4/1/2017	4/13/2017	47	M	U	E	Apr-17	13	...	0	0	0	0	0	0	0	0	0	0
8	9	67070	4/1/2017	4/3/2017	65	F	U	E	Apr-17	3	...	0	0	0	0	0	0	0	1	0	0
9	10	153218	4/1/2017	4/3/2017	59	M	U	E	Apr-17	3	...	0	0	0	0	0	0	0	0	0	0

10 rows × 56 columns

Date Parsing

```
In [91]: df['D.O.A'] = pd.to_datetime(df['D.O.A'], dayfirst=True, errors='coerce')
         df['D.O.D'] = pd.to_datetime(df['D.O.D'], dayfirst=True, errors='coerce')
```

3. Duration collumn convert into number

```
In [92]:
         df['DURATION OF STAY'] = pd.to_numeric(df['DURATION OF STAY'], errors='coerce')
```

```
In [93]: df_clean = df.dropna(subset=['D.O.A', 'D.O.D'])
         print(f"Final Data Size: {len(df_clean)}")
         print(f"Data Lost: {len(df) - len(df_clean)} (Much better than 5500!)")
```

```
Final Data Size: 11040
Data Lost: 4717 (Much better than 5500!)
```

occupancy (Time Series)

```
In [94]:
         start_date = df_clean['D.O.A'].min()
         end_date = df_clean['D.O.D'].max()
         all_dates = pd.date_range(start=start_date, end=end_date, freq='D')
         occupancy_list = []
         for current_date in all_dates:

             # (DOD > Date)
             active = ((df_clean['D.O.A'] <= current_date) & (df_clean['D.O.D'] > current_date)).sum()
             occupancy_list.append(active)
```

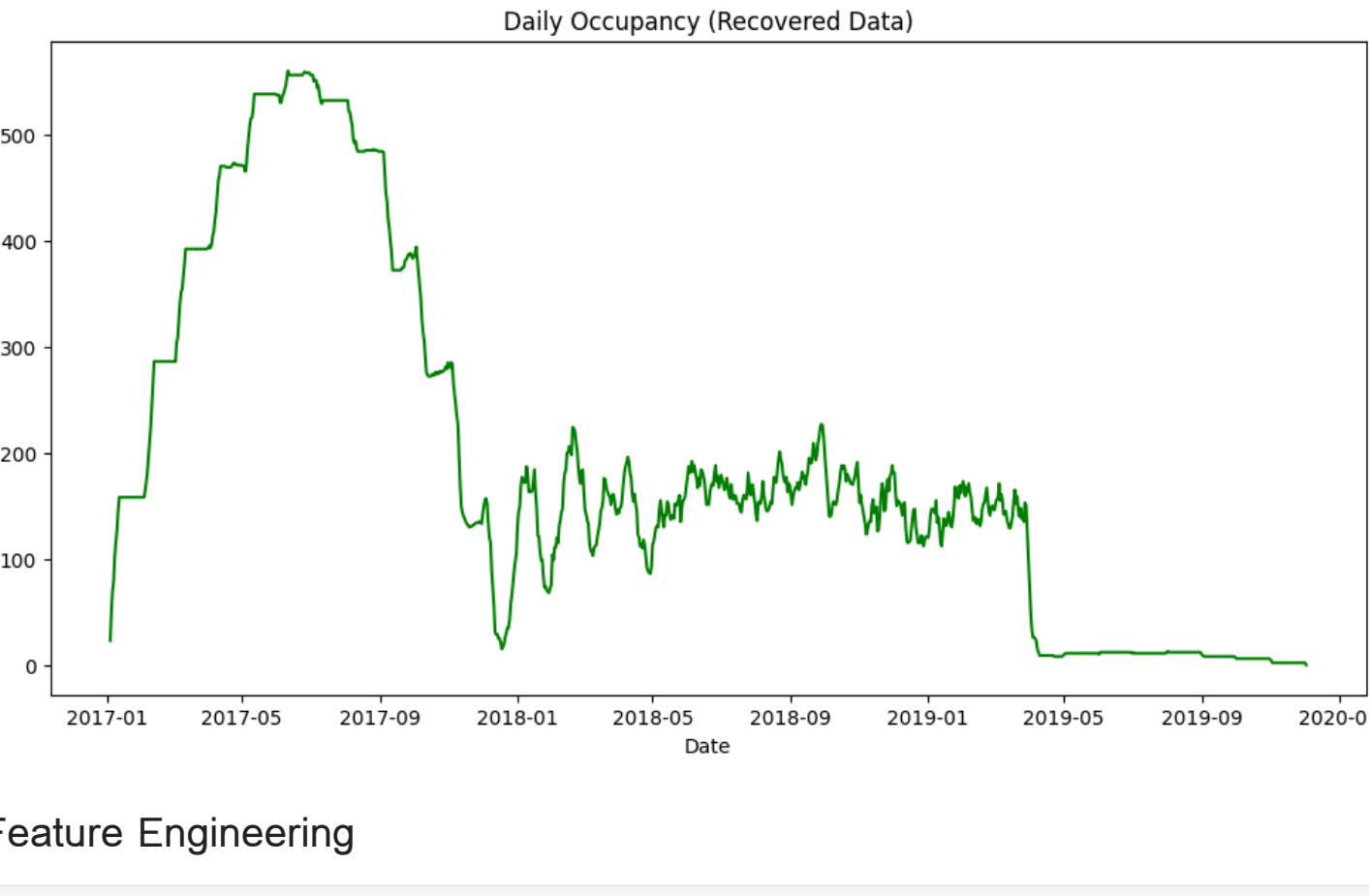
```
In [95]: df.isnull().sum()
```

```
Out[95]: SNO                                   0
         MRD No.                               0
         D.O.A                              3767
         D.O.D                              3770
         AGE                                   0
         GENDER                                0
         RURAL                                 0
         TYPE OF ADMISSION-EMERGENCY/OPD       0
         month year                            0
         DURATION OF STAY                      0
         duration of intensive unit stay       0
         OUTCOME                               0
         SMOKING                               0
         ALCOHOL                               0
         DM                                    0
         HTN                                   0
         CAD                                   0
         PRIOR CMP                             0
         CKD                                   0
         HB                                  252
         TLC                                 286
         PLATELETS                           285
         GLUCOSE                             863
         UREA                                241
         CREATININE                          247
         BNP                                8441
         RAISED CARDIAC ENZYMES                0
         EF                                 1505
         SEVERE ANAEMIA                        0
         ANAEMIA                               0
         STABLE ANGINA                         0
         ACS                                   0
         STEMI                                 0
         ATYPICAL CHEST PAIN                   0
         HEART FAILURE                         0
         HFREF                                 0
         HFNEF                                 0
         VALVULAR                              0
         CHB                                   0
         SSS                                   0
         AKI                                   0
         CVA INFRACT                           0
         CVA BLEED                             0
         AF                                    0
         VT                                    0
         PSVT                                  0
         CONGENITAL                            0
         UTI                                   0
         NEURO CARDIOGENIC SYNCOPE             0
         ORTHOSTATIC                           0
         INFECTIVE ENDOCARDITIS                0
         DVT                                   0
         CARDIOGENIC SHOCK                     0
         SHOCK                                 0
         PULMONARY EMBOLISM                    0
         CHEST INFECTION                       0
         dtype: int64
```

```
In [96]:
         final_df = pd.DataFrame({'Date': all_dates, 'Total_Occupancy': occupancy_list})
         final_df.to_csv('processed_occupancy_v3.csv', index=False)
```

visulization

```
In [97]:
         plt.figure(figsize=(12, 6))
         sns.lineplot(data=final_df, x='Date', y='Total_Occupancy', color='green')
         plt.title('Daily Occupancy (Recovered Data)')
         plt.show()
```



Feature Engineering

```
In [98]: df = pd.read_csv('processed_occupancy_v3.csv')
         df['Date'] = pd.to_datetime(df['Date'])
```

calender

```
In [99]: df['Day_of_Week'] = df['Date'].dt.dayofweek

         # 1 = January, 12 = December
         df['Month'] = df['Date'].dt.month

         #  (1-31)
         df['Day_of_Month'] = df['Date'].dt.day
```

before 1 and 7 days paitent pattern

```
In [100… df['Lag_1'] = df['Total_Occupancy'].shift(1)

         df['Lag_7'] = df['Total_Occupancy'].shift(7)
```

last 7 days paitent mean

```
In [101…
         df['Rolling_Mean_7'] = df['Total_Occupancy'].rolling(window=7).mean()
```

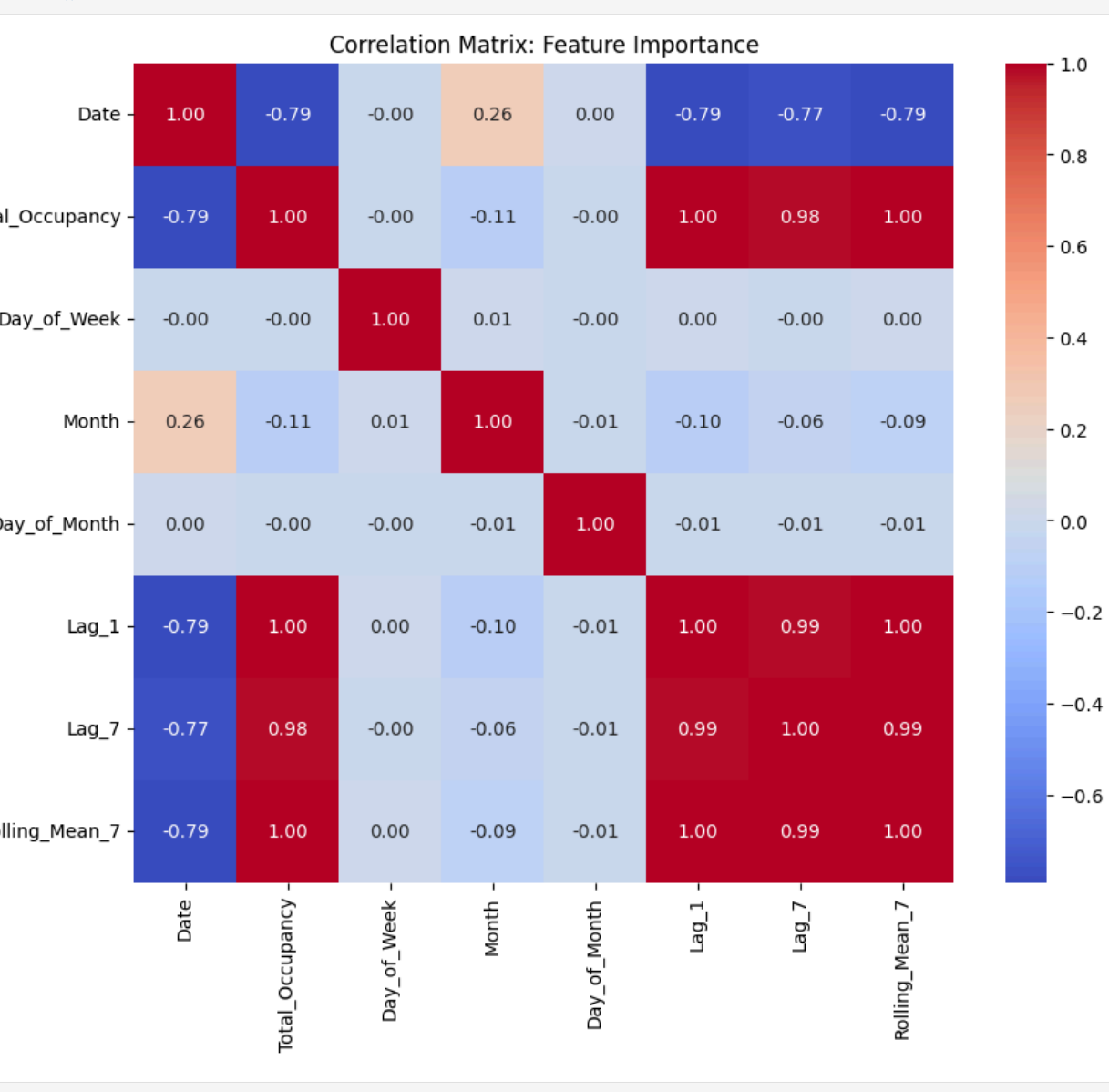
```
In [102…
         df_ml = df.dropna()
```

```
In [103…

         print(df_ml.head())
```

```
         Date  Total_Occupancy  Day_of_Week  Month  Day_of_Month  Lag_1  \
7  2017-01-11              145            2      1            11  127.0   
8  2017-01-12              158            3      1            12  145.0   
9  2017-01-13              158            4      1            13  158.0   
10 2017-01-14              158            5      1            14  158.0   
11 2017-01-15              158            6      1            15  158.0   

    Lag_7  Rolling_Mean_7  
7    23.0       98.285714  
8    50.0      113.714286  
9    69.0      126.428571  
10   79.0      137.714286  
11  102.0      145.714286  
```

Correlation Heatmap

```
In [104…
         plt.figure(figsize=(10, 8))
         sns.heatmap(df_ml.corr(), annot=True, cmap='coolwarm', fmt='.2f' ,square=True)
         plt.title('Correlation Matrix: Feature Importance')
         plt.show()
```



```
In [105… df_ml.to_csv('final_ml_data.csv', index=False)
         print('final_ml_data.csv'  )
```

```
final_ml_data.csv
```

Machine learning part

Random forest

```
In [106… df=pd.read_csv('final_ml_data.csv')
```

```
In [107… df.head()
```

Out[107…

	Date	Total_Occupancy	Day_of_Week	Month	Day_of_Month	Lag_1	Lag_7	Rolling_Mean_7
0	2017-01-11	145	2	1	11	127.0	23.0	98.285714
1	2017-01-12	158	3	1	12	145.0	50.0	113.714286
2	2017-01-13	158	4	1	13	158.0	69.0	126.428571
3	2017-01-14	158	5	1	14	158.0	79.0	137.714286
4	2017-01-15	158	6	1	15	158.0	102.0	145.714286

```
In [108… x = df.drop(['Total_Occupancy', 'Date'], axis=1)

         y = df['Total_Occupancy']
```

```
In [109… from sklearn.model_selection import train_test_split
         x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
```

```
In [110… from sklearn.ensemble import RandomForestRegressor
         model = RandomForestRegressor(n_estimators=100, random_state=42)
         model.fit(x_train, y_train)
```

```
Out[110… ▾ RandomForestRegressor
         ▸ Parameters
```

```
In [111… y_pred = model.predict(x_test)
         from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
         mae = mean_absolute_error(y_test, y_pred)
         mse = mean_squared_error(y_test, y_pred)
         r2 = r2_score(y_test, y_pred)
```

```
In [112…

         print(" Result of Random Forest ")
         print(f"Accuracy (R² Score): {r2:.2f}")
         print(f"Average Error (MAE): {mae:.2f}")
         print(f"Mean Squared Error (MSE): {mse:.2f}")
```

```
 Result of Random Forest
Accuracy (R² Score): 1.00
Average Error (MAE): 5.23
Mean Squared Error (MSE): 80.75
```

linear Regression

```
In [113… from sklearn.linear_model import  LinearRegression
```

```
In [114… model= LinearRegression()
         model.fit(x_train, y_train)
```

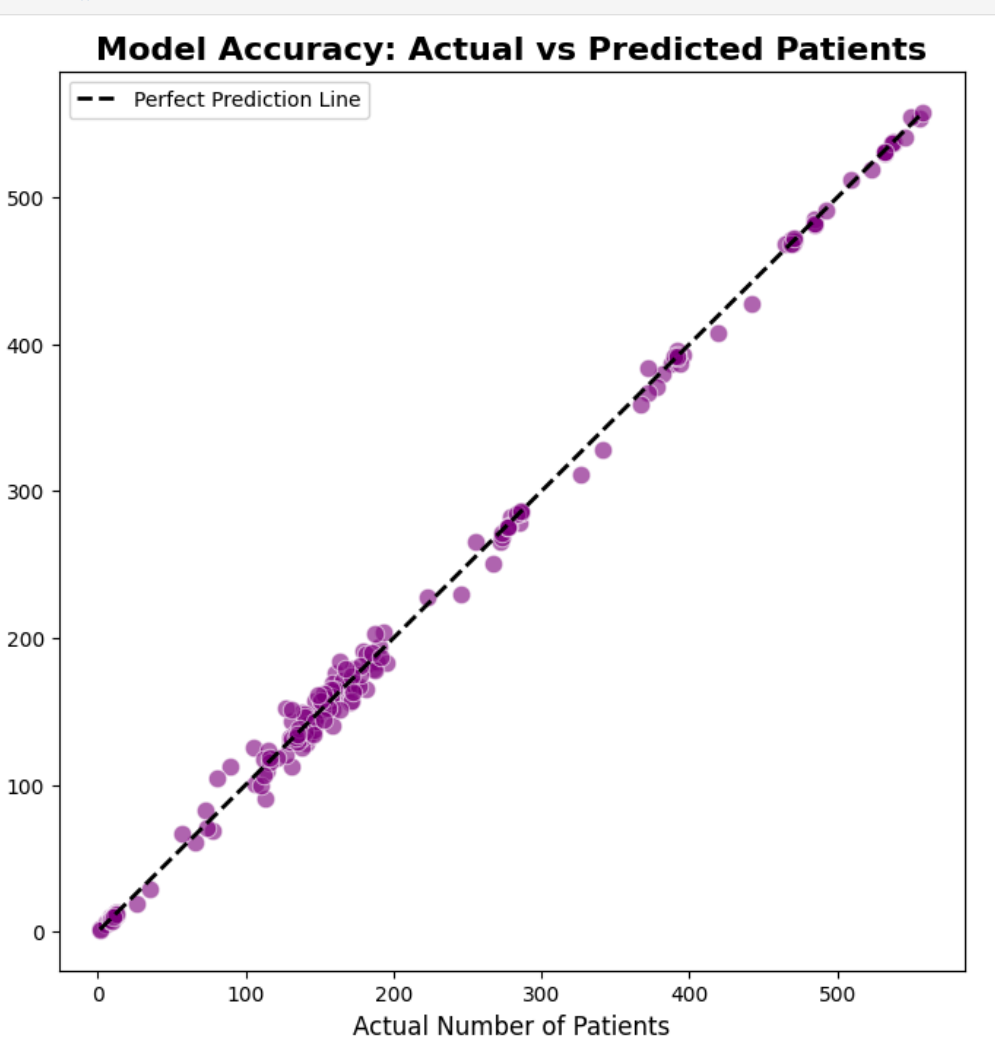
```
Out[114… ▾ LinearRegression
         ▸ Parameters
```

```
In [115… y_pred= model.predict(x_test)
         mae = mean_absolute_error(y_test, y_pred)
         mse = mean_squared_error(y_test, y_pred)
         r2 = r2_score(y_test, y_pred)
```

```
In [116… print(" Result of Linear Regression ")
         print(f"Accuracy (R² Score): {r2:.2f}")
         print(f"Average Error (MAE): {mae:.2f}")
         print(f"Mean Squared Error (MSE): {mse:.2f}")
```

```
 Result of Linear Regression
Accuracy (R² Score): 1.00
Average Error (MAE): 4.93
Mean Squared Error (MSE): 53.91
```

```
In [117…

         plt.figure(figsize=(8, 8))

         sns.scatterplot(x=y_test, y=y_pred, color='purple', alpha=0.6, s=80)


         max_val = max(max(y_test), max(y_pred))
         min_val = min(min(y_test), min(y_pred))
         plt.plot([min_val, max_val], [min_val, max_val], color='black', linestyle='--', linewidth=2, label='Perfect Prediction Line')

         plt.title('Model Accuracy: Actual vs Predicted Patients', fontsize=16, fontweight='bold')
         plt.xlabel('Actual Number of Patients', fontsize=12)
         plt.ylabel('AI Predicted Number of Patients', fontsize=12)
         plt.legend()
         plt.show()
```



GradientBoostingRegressor

```
In [118… from sklearn.ensemble import GradientBoostingRegressor
```

```
In [119… gb_model = GradientBoostingRegressor(n_estimators=100, random_state=42)
         gb_model.fit(x_train, y_train)
```

```
Out[119… ▾ GradientBoostingRegressor
         ▸ Parameters
```

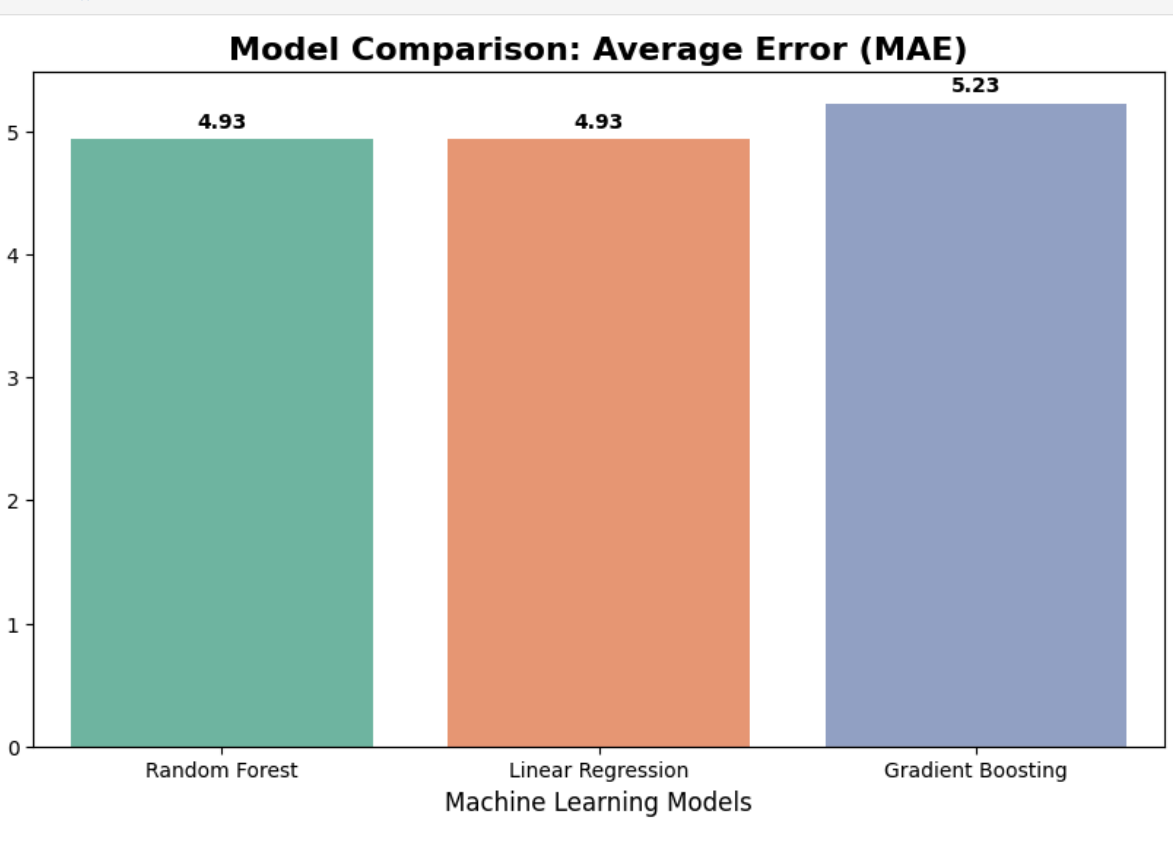
```
In [120… y_pred_gb = gb_model.predict(x_test)
         mae_gb = mean_absolute_error(y_test, y_pred_gb)
         mse_gb = mean_squared_error(y_test, y_pred_gb)
         r2_gb = r2_score(y_test, y_pred_gb)
```

```
In [121… print(" Result of Gradient Boosting ")

         print(f"Accuracy (R² Score): {r2_gb:.2f}")
         print(f"Average Error (MAE): {mae_gb:.2f}")
         print(f"Mean Squared Error (MSE): {mse_gb:.2f}")
```

```
 Result of Gradient Boosting
Accuracy (R² Score): 1.00
Average Error (MAE): 5.23
Mean Squared Error (MSE): 67.50
```

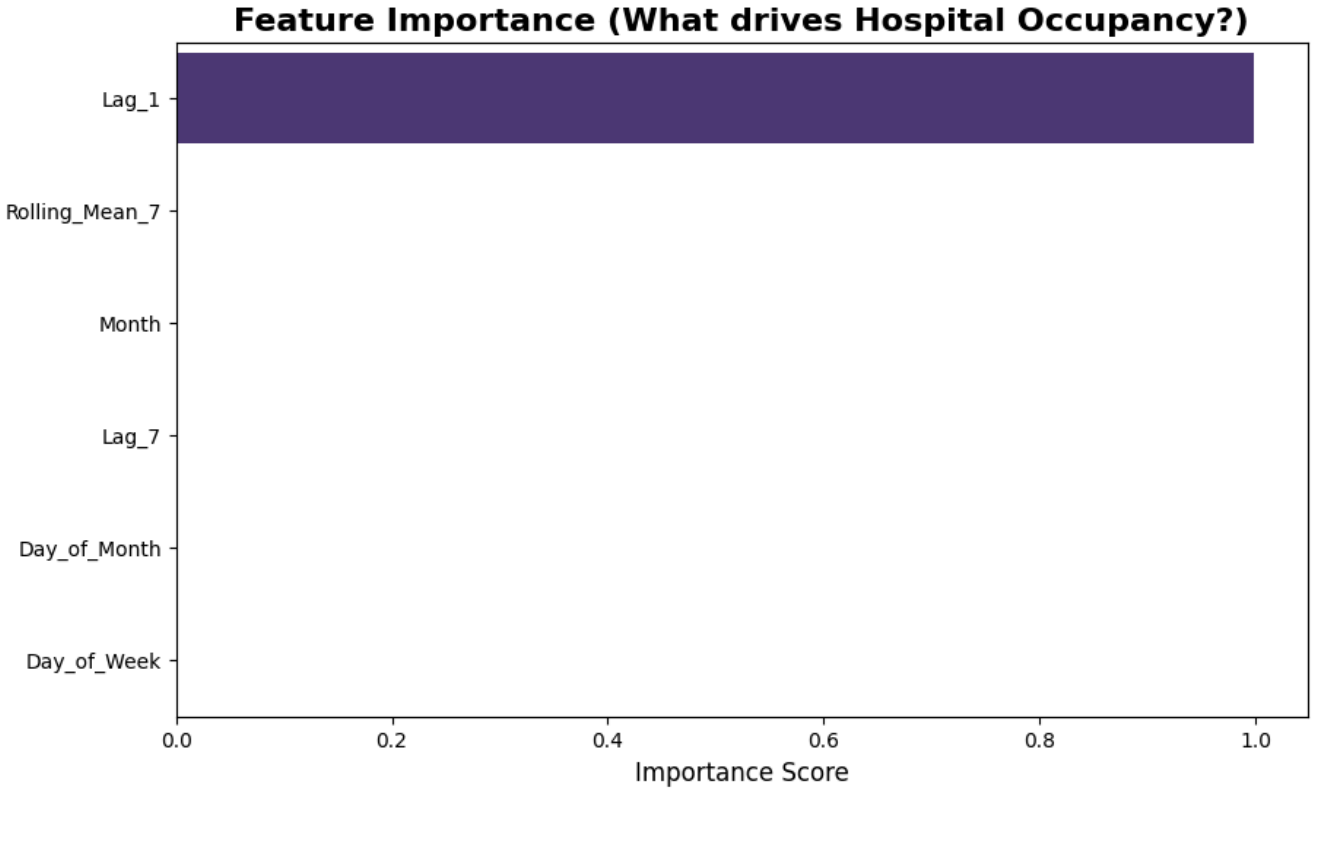
Model Comparison: Which model performed best

```
In [122…

         import matplotlib.pyplot as plt
         import seaborn as sns
         models = ['Random Forest', 'Linear Regression', 'Gradient Boosting']
         mae_scores = [mae, mean_absolute_error(y_test, model.predict(x_test)), mae_gb]
         plt.figure(figsize=(10, 6))
         sns.barplot(x=models, y=mae_scores, palette='Set2')
         plt.title('Model Comparison: Average Error (MAE)', fontsize=16, fontweight='bold')
         plt.xlabel('Machine Learning Models', fontsize=12)
         plt.ylabel('Mean Absolute Error (Lower is Better)', fontsize=12)

         for i, v in enumerate(mae_scores):
             plt.text(i, v + 0.1, f"{v:.2f}", color='black', ha='center', fontweight='bold')

         plt.show()
```



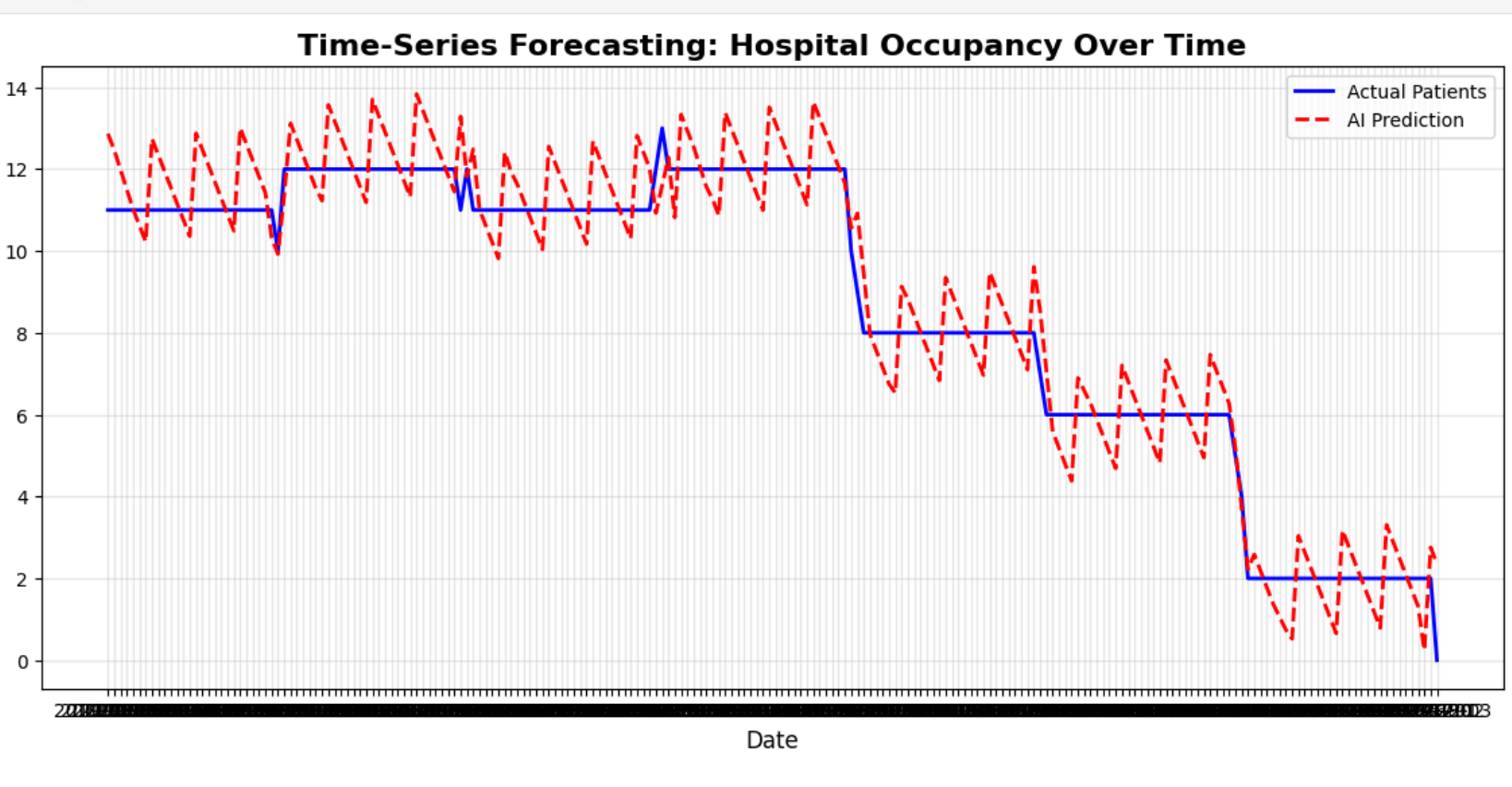
Feature Importance: Which clues helped the AI most?

```
In [123…

         import numpy as np

         feature_importance = gb_model.feature_importances_
         feature_names = x.columns

         indices = np.argsort(feature_importance)[::-1]

         plt.figure(figsize=(10, 6))
         sns.barplot(x=feature_importance[indices], y=feature_names[indices], palette='viridis')

         plt.title('Feature Importance (What drives Hospital Occupancy?)', fontsize=16, fontweight='bold')
         plt.xlabel('Importance Score', fontsize=12)
         plt.ylabel('Features (Clues)', fontsize=12)
         plt.show()
```



Real-time Trend: Actual vs Predicted over Time

```
In [124…

         split_idx = int(len(x) * 0.8)
         x_train_seq, x_test_seq = x.iloc[:split_idx], x.iloc[split_idx:]
         y_train_seq, y_test_seq = y.iloc[:split_idx], y.iloc[split_idx:]
         dates_test = df['Date'].iloc[split_idx:]


         best_model.fit(x_train_seq, y_train_seq)
         y_pred_seq = best_model.predict(x_test_seq)

         plt.figure(figsize=(14, 6))

         plt.plot(dates_test, y_test_seq, label='Actual Patients', color='blue', linewidth=2)

         plt.plot(dates_test, y_pred_seq, label='AI Prediction', color='red', linestyle='--', linewidth=2)

         plt.title('Time-Series Forecasting: Hospital Occupancy Over Time', fontsize=16, fontweight='bold')
         plt.xlabel('Date', fontsize=12)
         plt.ylabel('Number of Patients', fontsize=12)
         plt.legend(loc='upper right')
         plt.grid(True, alpha=0.3)
         plt.show()
```



🚨 AI-Based Early Warning System Dashboard

```
In [125…

         #  use for last row of data for prediction
         latest_data = x_test_seq.iloc[-1:]

         #  convert string to datetime and format it
         latest_date_raw = dates_test.iloc[-1]
         latest_date = pd.to_datetime(latest_date_raw).strftime('%d %B, %Y')

         # use our best model (Linear Regression) to predict the number of patients for the next day
         predicted_occupancy = int(best_model.predict(latest_data)[0])

         #  use for sample hospital bed capacity and alert threshold
         TOTAL_BEDS = 500
         WARNING_THRESHOLD = 450

         print("=" * 55)
         print("🏥 HOSPITAL EARLY WARNING DASHBOARD 🏥")
         print("=" * 55)
         print(f"📅 Date (Forecast)  : {latest_date}")
         print(f"🛏 Total Capacity   : {TOTAL_BEDS} Beds")
         print(f"📊 Expected Patients: {predicted_occupancy} Patients")
         print("-" * 55)

         #  AI Alert Logic
         if predicted_occupancy >= TOTAL_BEDS:
             print("🚨 [CRITICAL ALERT]: Hospital is at OVERCAPACITY!")
             print("👉 Action: Stop new regular admissions. Prepare emergency floor beds.")

         elif predicted_occupancy >= WARNING_THRESHOLD:
             print("⚠️ [WARNING]: High patient volume detected! (Over 90% Booked)")
             print("👉 Action: Keep emergency staff on standby. Fast-track patient discharges.")

         else:
             print("✅ [STATUS NORMAL]: Sufficient beds are available.")
             print("👉 Action: Continue normal hospital operations.")

         print("=" * 55)
```

```
=======================================================
🏥 HOSPITAL EARLY WARNING DASHBOARD 🏥
=======================================================
📅 Date (Forecast)  : 03 December, 2019
🛏 Total Capacity   : 500 Beds
📊 Expected Patients: 2 Patients
-------------------------------------------------------
✅ [STATUS NORMAL]: Sufficient beds are available.
👉 Action: Continue normal hospital operations.
=======================================================
```

```
In [126…

         row_number = -10

         latest_data = x_test_seq.iloc[row_number : row_number+1].copy()
         latest_date_raw = dates_test.iloc[row_number]
         latest_date = pd.to_datetime(latest_date_raw).strftime('%d %B, %Y')


         latest_data['Lag_1'] = 465
         latest_data['Rolling_Mean_7'] = 460


         predicted_occupancy = int(best_model.predict(latest_data)[0])

         TOTAL_BEDS = 500
         WARNING_THRESHOLD = 450

         print("=" * 55)
         print("🏥 HOSPITAL EARLY WARNING DASHBOARD 🏥")
         print("=" * 55)
         print(f"📅 Date (Forecast)  : {latest_date}")
         print(f"🛏 Total Capacity   : {TOTAL_BEDS} Beds")
         print(f"📊 Expected Patients: {predicted_occupancy} Patients")
         print("-" * 55)

         if predicted_occupancy >= TOTAL_BEDS:
             print("🚨 [CRITICAL ALERT]: Hospital is at OVERCAPACITY!")
             print("👉 Action: Stop new regular admissions. Prepare emergency floor beds.")

         elif predicted_occupancy >= WARNING_THRESHOLD:
             print("⚠️ [WARNING]: High patient volume detected! (Over 90% Booked)")
             print("👉 Action: Keep emergency staff on standby. Fast-track patient discharges.")

         else:
             print("✅ [STATUS NORMAL]: Sufficient beds are available.")
             print("👉 Action: Continue normal hospital operations.")

         print("=" * 55)
```

```
=======================================================
🏥 HOSPITAL EARLY WARNING DASHBOARD 🏥
=======================================================
📅 Date (Forecast)  : 24 November, 2019
🛏 Total Capacity   : 500 Beds
📊 Expected Patients: 503 Patients
-------------------------------------------------------
🚨 [CRITICAL ALERT]: Hospital is at OVERCAPACITY!
```

```
👉 Action: Stop new regular admissions. Prepare emergency floor beds.
=======================================================
```